

Variable		Description	1-prong	3-prong
Track inputs	$p_{\text{T}}^{\tau_{\text{had-vis}}}$	Transverse momentum of the $\tau_{\text{had-vis}}$ candidate	✓	✓
	$p_{\text{T}}^{\text{track}}$	Transverse momentum of each associated track	✓	✓
	$\Delta\eta^{\text{track}}$	Angular distance between the track and $\tau_{\text{had-vis}}$ axis	✓	✓
	$\Delta\phi^{\text{track}}$	Angular distance between the track and $\tau_{\text{had-vis}}$ axis	✓	✓
	N_{IBL}	Number of hits in the innermost layer of the pixel detector	✓	✓
	N_{pixel}	Number of hits in the pixel detector	✓	✓
	N_{SCT}	Number of hits in the SCT	✓	✓
	$z_0^{\text{TJVA}} \sin\theta$	Tau vertex longitudinal impact parameter	✓	✓
	d_0^{TJVA}	Tau vertex transverse impact parameter	✓	✓
Cluster inputs	$S(z_0^{\text{TJVA}} \sin\theta)$	Tau vertex longitudinal impact parameter divided by its uncertainty	✓	✓
	$S(d_0^{\text{TJVA}})$	Tau vertex transverse impact parameter divided by its uncertainty	✓	✓
	$p_{\text{T}}^{\tau_{\text{had-vis}}}$	Transverse momentum of the $\tau_{\text{had-vis}}$ candidate	✓	✓
	$E_{\text{T}}^{\text{cluster}}$	TopoCluster transverse energy	✓	✓
	$\Delta\eta^{\text{cluster}}$	The angular distance between the TopoCluster and the $\tau_{\text{had-vis}}$ axis	✓	✓
	$\Delta\phi^{\text{cluster}}$	The angular distance between the TopoCluster and the $\tau_{\text{had-vis}}$ axis	✓	✓
	$\langle\lambda_{\text{centre}}\rangle$	Mean distance of the cluster shower centre from the calorimeter front face measured along the shower axis.	✓	✓
	$\langle\lambda^2\rangle$	Mean distance of a cell from the shower centre along the shower axis	✓	✓
	$\langle r^2\rangle$	Second moment in the radial distance of cluster cells from the cluster axis.	✓	✓
High-level inputs	$p_{\text{T}}^{\tau_{\text{had-vis}}}$	Transverse momentum of the $\tau_{\text{had-vis}}$ candidate	✓	✓
	$f_{\text{cent}}^{\Delta R < 0.1}$	Fraction of transverse energy at EM scale deposited in the region $\Delta R < 0.1$ with respect to all energy deposited in $\Delta R < 0.2$ around the $\tau_{\text{had-vis}}$ candidate axis	✓	✓
	$f_{\text{lead-track}}^{-1}$	The transverse energy sum, calibrated at the EM energy scale, deposited in all cells belonging to TopoClusters in the core region, divided by the transverse momentum of the highest- p_{T} core track of the $\tau_{\text{had-vis}}$ candidate.	✓	✓
	$\Delta R_{\text{max}}(\text{track}, \tau_{\text{had}})$	The maximum ΔR between a core track associated with the $\tau_{\text{had-vis}}$ candidate and the $\tau_{\text{had-vis}}$ direction.	✓	✓
	$S_{\text{T}}^{\text{flight}}$	The decay length of the secondary vertex in the transverse plane, calculated with respect to the τ vertex, divided by its estimated uncertainty		✓
	$f_{\text{track}}^{\text{iso}}$	Scalar sum of the p_{T} of isolation tracks associated with the $\tau_{\text{had-vis}}$ candidate divided by the sum of the pT of all the associated core and isolation tracks.	✓	✓
	$f_{\text{track}}^{\text{EM}}$	Ratio of the sum of cluster energy deposited in the electromagnetic part of the TopoClusters (pre-sampler, first and second layers of the LAr calorimeter) to the sum of the momentum of core tracks.	✓	✓
	$p_{\text{T}}^{\text{EM+track}}/p_{\text{T}}$	Ratio of the $\tau_{\text{had-vis}}$ p_{T} , estimated using the vector sum of core track momenta and up to two most energetic EM clusters in the core region to the calorimeter-only measurement of $\tau_{\text{had-vis}}$ p_{T} .	✓	✓
	$m^{\text{EM+track}}$	Invariant mass of the system composed of the core tracks and up to two most energetic EM clusters in the core region and the four-momentum of an EM cluster is calculated assuming zero mass and using TopoCluster seed direction.	✓	✓
	m^{track}	Invariant mass calculated from the sum of the four-momenta of all core and isolation tracks, assuming a pion mass for each track		✓